Design And Implementation of a Simulated WISP Network Using MikroTik And Cambium PtMP Architecture

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Executive Summary

This lab demonstrates the design and implementation of a simulated Wireless Internet Service Provider (WISP) topology using a mobile hotspot as the upstream internet source, a MikroTik router as the edge gateway, and Cambium Point-to-Multipoint (PtMP) radios for last-mile distribution.

The MikroTik router was configured to operate as a Wireless WAN (WWAN) client connecting to a mobile hotspot. It performed routing, Network Address Translation (NAT), and DHCP services for downstream devices. A Cambium Access Point (AP) was deployed as the base station, distributing connectivity to two Subscriber Modules (SM1 and SM2), representing customer premises equipment (CPEs).

This architecture simulates a real-world ISP deployment model using scaled-down lab equipment and infrastructure.

1.0 Introduction

Wireless Internet Service Providers (WISPs) commonly deploy Point-to-Multipoint (PtMP) architectures to distribute internet connectivity to multiple customers from a centralized base station. This lab recreates such an environment using mobile hotspot as a simulated upstream core connection, MikroTik router acting as an edge gateway, Cambium radios operating in AP and Subscriber modes

The lab focuses on understanding traffic flow, gateway routing, NAT operation, DHCP distribution, and wireless distribution over a radio access network.

2.0 Aim and Objectives

The aim of this lab is to design and implement a functional simulated WISP network using MikroTik routing and Cambium PtMP radios. The objectives includes;

- Configuring MikroTik as a wireless WAN (WWAN) client
- Implementing NAT masquerading for internet sharing
- Deploying DHCP services for downstream devices
- Configuring Cambium AP in Base Station mode
- Configuring Subscriber Modules (SMs) to associate with the AP
- Validating end-to-end internet connectivity
- Analyzing performance limitations of double wireless hops

3.0 Tools and Equipment

- MikroTik Router (RB951Ui-2HnD)
- Cambium Networks PtMP Radios
- Mobile phone hotspot (4G/5G network)
- Ethernet cables
- PoE injector
- WinBox / Web GUI for configuration
- Laptop for configuration and testing

3.1 Devices and Roles

Device	Function in Lab	Real ISP Equivalent
Phone Hotspot	Upstream Internet Source	Fiber / Microwave Backhaul
MikroTik Router	WWAN Client + NAT + DHCP Gateway	Edge Router / BNG
Cambium AP	Base Station (PtMP)	Sector Antenna
SM1 & SM2	Subscriber Modules (CPE)	Customer Premises Equipment

4.0 Network Architecture Overview

Logical Traffic Flow
Mobile Network
↓
Phone Hotspot (Wi-Fi)
↓
MikroTik Router (Station Mode - WWAN)
↓
Cambium AP (Base Station)
↓ ↓
SM1 (CPE) SM2 (CPE)

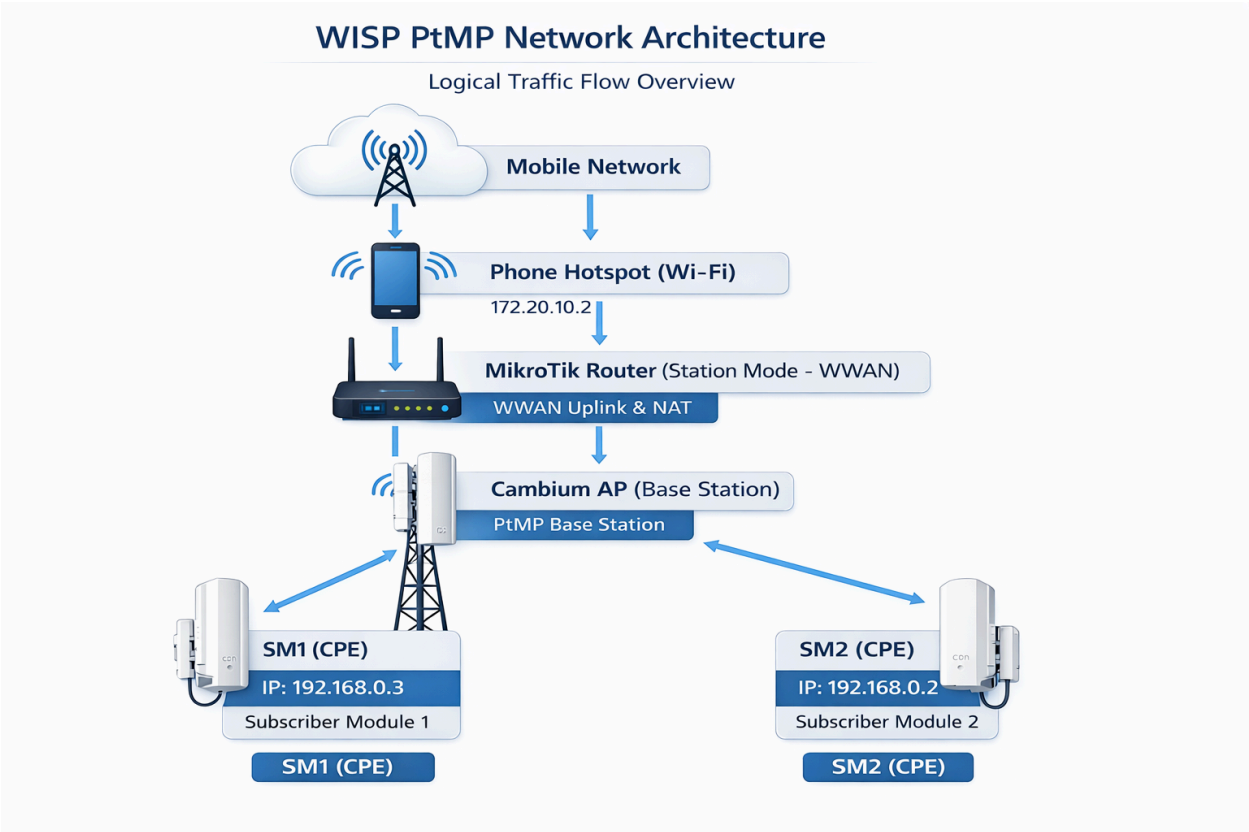

Fig 1: A network architectural design of the lab topology

4.1 IP Addressing Plan

Segment	Device	Interface	IP Address	Subnet Mask	Default Gateway	Assignment Type	Role
Upstream (WWAN)	Phone Hotspot	LAN	172.20.10.1	/28	N/A	Default	Internet Gateway
Upstream (WWAN)	MikroTik Router	wlan1	172.20.10.2	/28	172.20.10.1	DHCP Client	WAN Interface
Downstream (LAN)	MikroTik Router	ether1	192.168.0.254	/24	N/A	Static	LAN Gateway
Downstream (LAN)	Cambium AP (Base Station)	Management	192.168.0.1	/24	192.168.0.254	Static	PtMP Base Station
Downstream (LAN)	SM2 (CPE)	Management	192.168.0.2	/24	192.168.0.254	Static	Subscriber Module
Downstream (LAN)	SM1 (CPE)	Management	192.168.0.3	/24	192.168.0.254	Static	Subscriber Module

Downstream (LAN)	Client Devices (Optional)	DHCP	192.168.0.100–200	/24	192.168.0.254	DHCP Server	End Users

5.0 Methodology

The lab was implemented in structured phases:

Phase One: Wireless WAN Configuration on MikroTik

- wlan1 configured in **station mode** to function as a wireless client
- Connected to the mobile hotspot SSID
- WPA2 security parameters properly configured
- Internet connectivity verified using ICMP ping tests

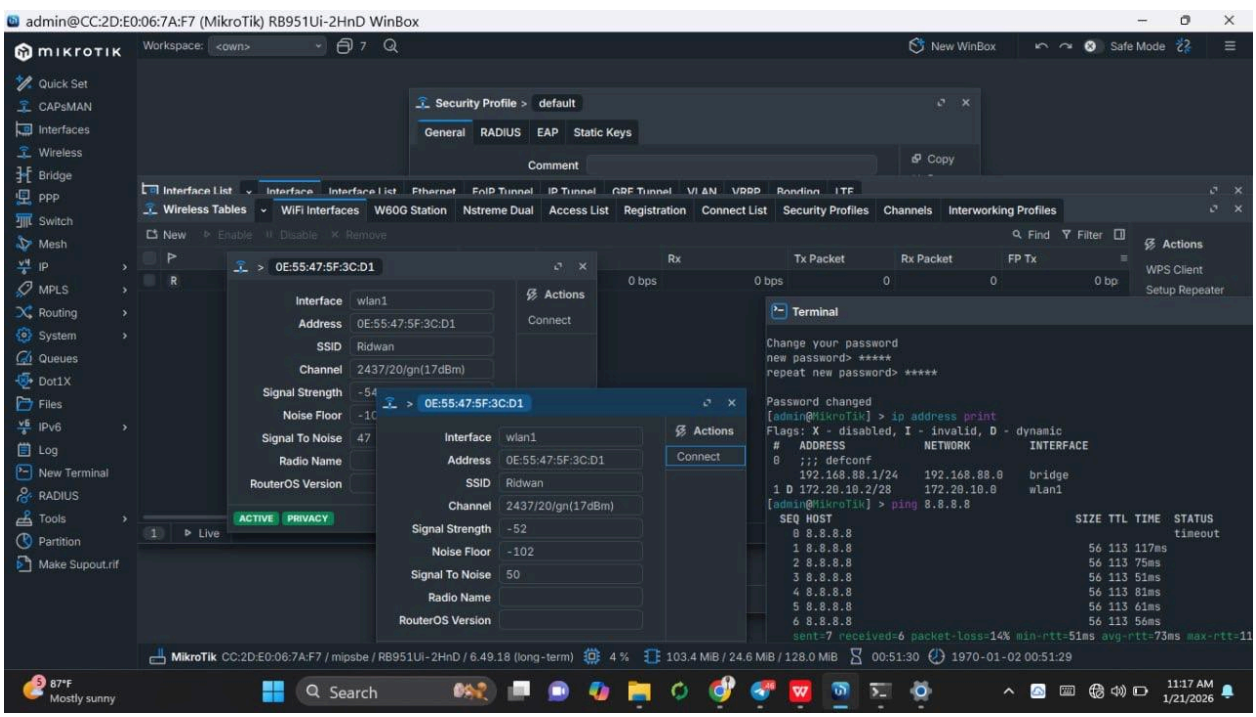

Fig 2: Evidence from winbox interface showing a successful wireless connection of the mikrotik router to the hotspot

Phase Two: LAN and NAT Configuration

- Assigned static IP address **192.168.0.254 /24** to ether1 (LAN interface)
- Created DHCP server for the **192.168.0.100-200 /24** downstream network
- Configured NAT masquerade rule (Out-interface: wlan1)
- Verified routing table and default gateway functionality

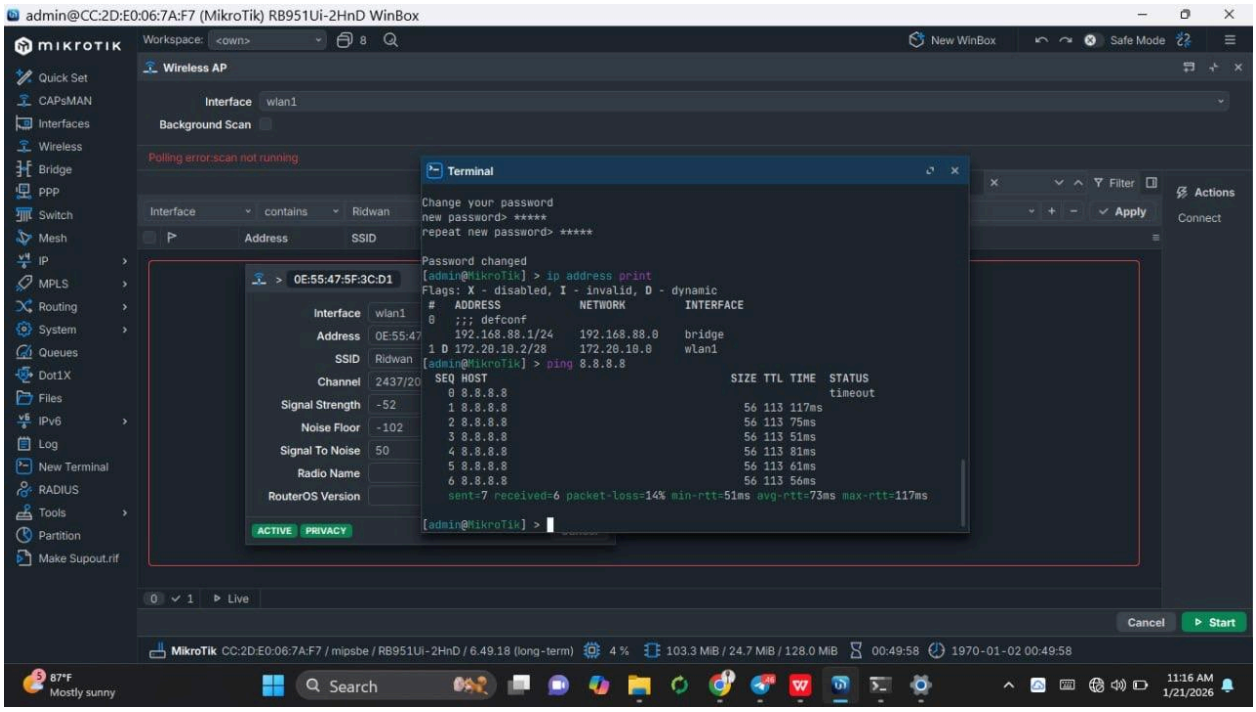

Fig 3: Evidence from winbox interface showing wlan ip address on the mikrotik router.

Phase Three: Cambium AP Configuration

- Mode configured as **Access Point (PtMP Base Station)**
- Management IP address set to **192.168.0.1 /24**
- Default gateway configured as **192.168.0.254 /24**
- DHCP service disabled (centrally handled by MikroTik)
- SSID configured to enable subscriber module association

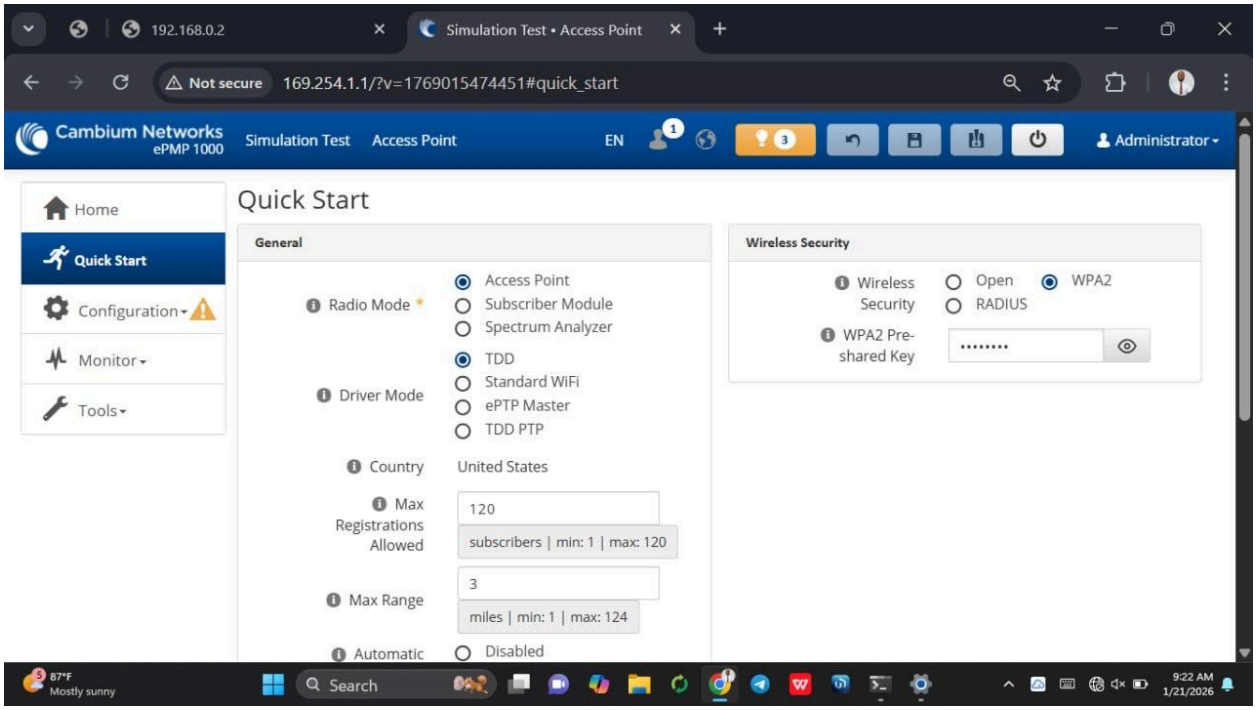

Fig 4: Evidence showing the cambium Radio PtMP Access Point interface

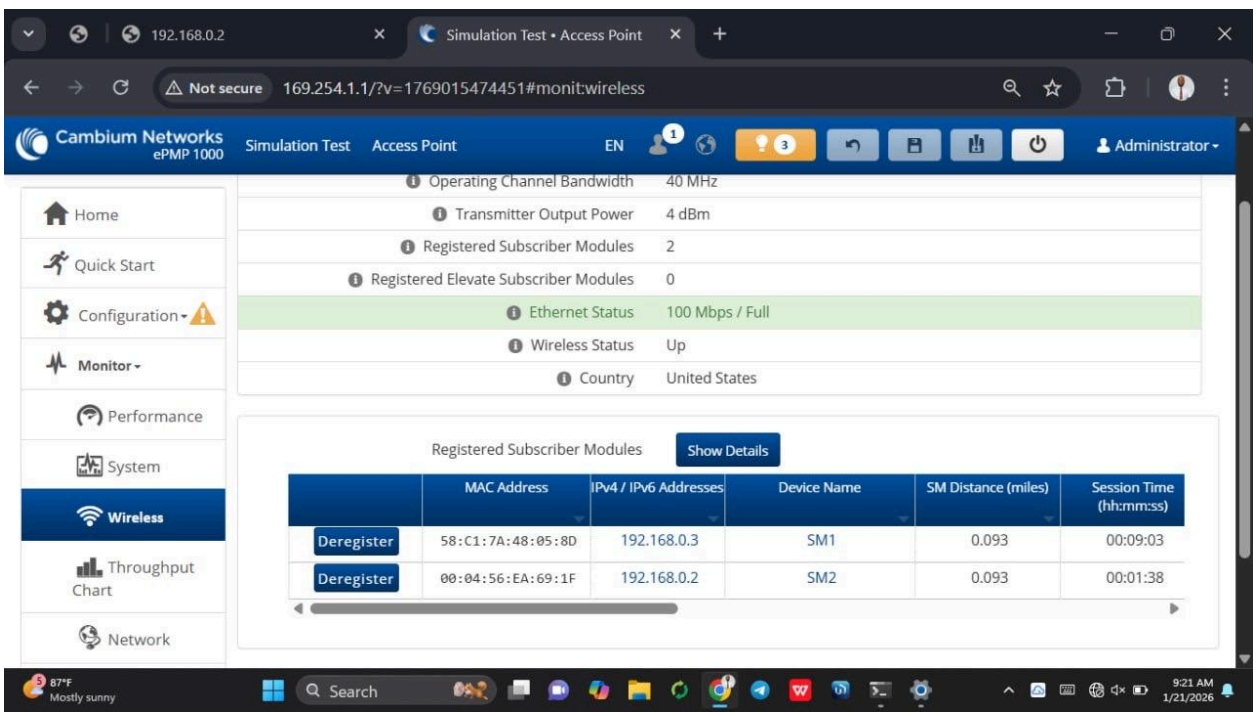

Fig 5: Evidence showing the two SM successfully registered on the AP

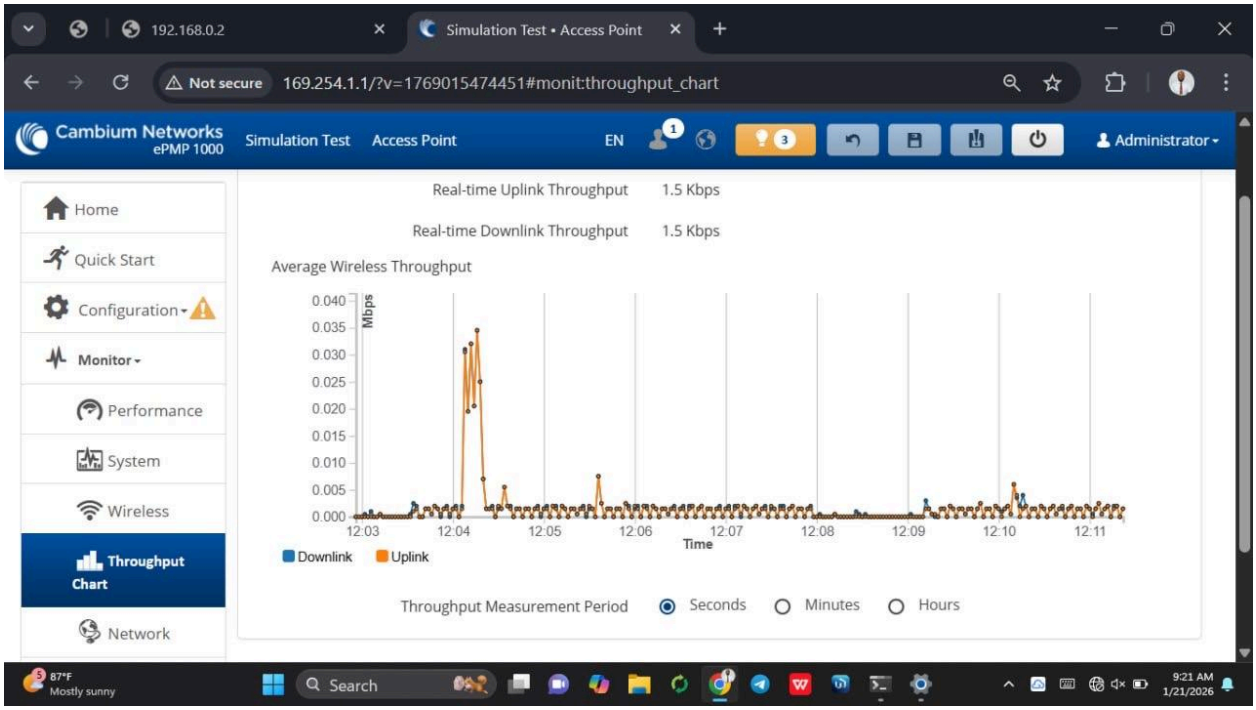

Fig 6: Evidence showing the throughput chat on the AP

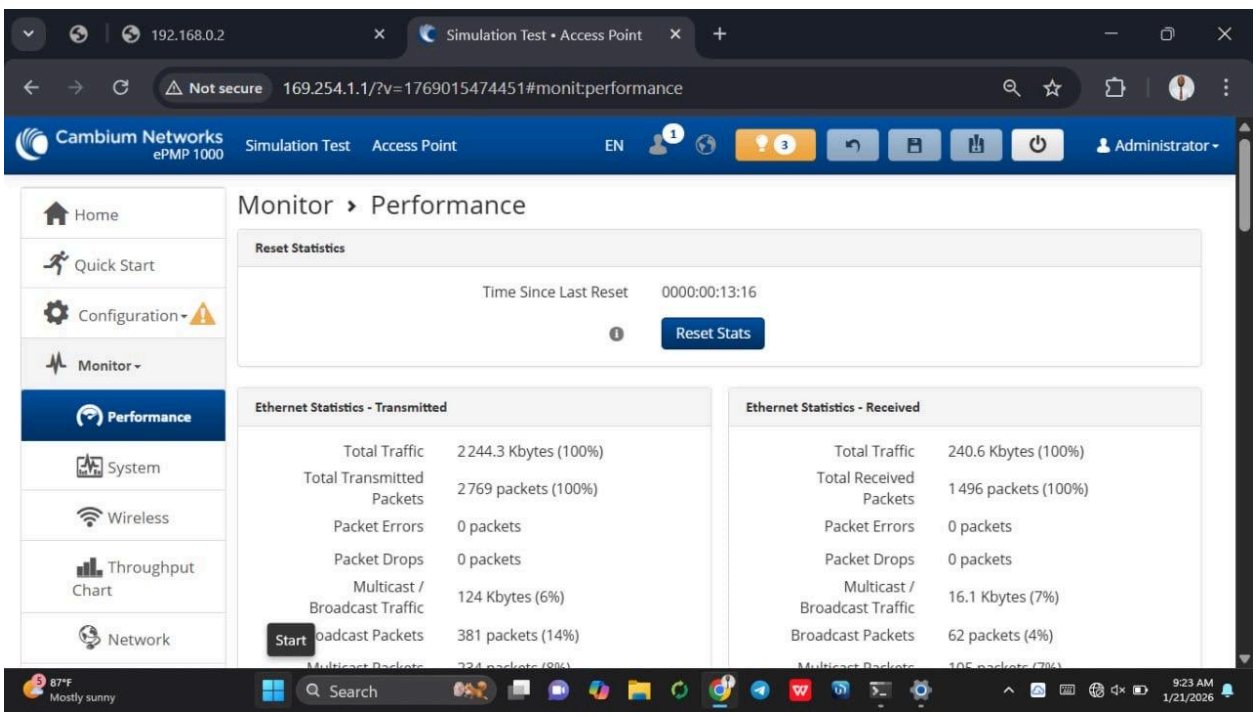

Fig 7: Evidence showing the performance information on the AP Radio Interface

Phase Four: Subscriber Module Configuration

SM1 Configuration

- Mode set to **Subscriber**
- Management IP address: **192.168.0.3 /24**
- Default Gateway: **192.168.0.254 /24**
- Successfully associated with the Access Point and established a stable wireless link

SM2 Configuration

- Mode set to **Subscriber**
- Management IP address: **192.168.0.2 /24**
- Default Gateway: **192.168.0.254 /24**
- Successfully associated with the Access Point and established a stable wireless link

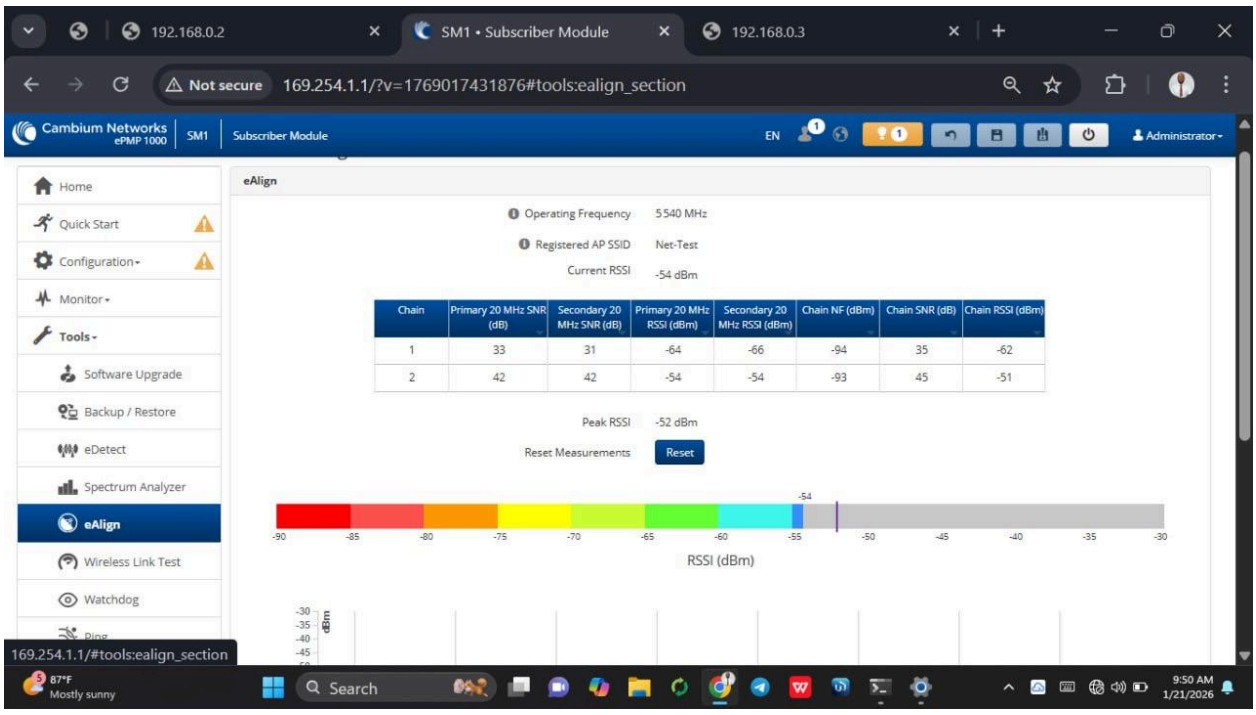

Fig 8: Evidence showing a strong RSSI from the SM interface

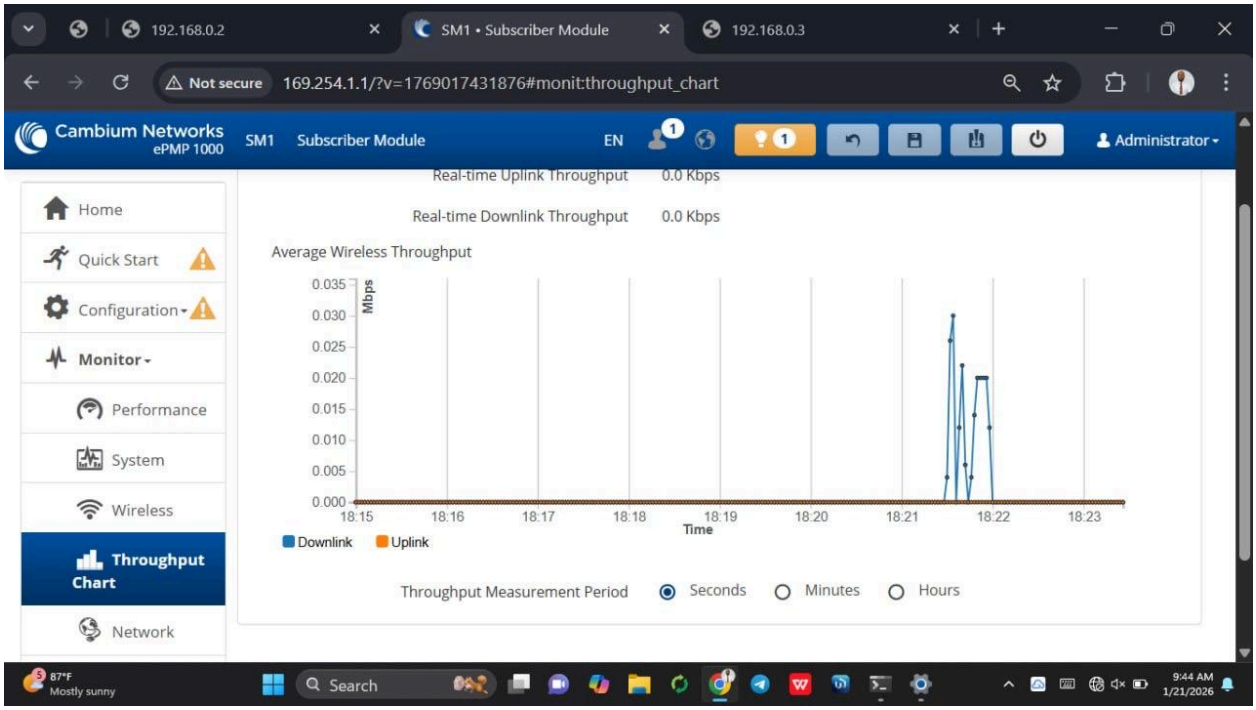

Fig 9: Evidence showing the throughput chat from the DM Radio interface

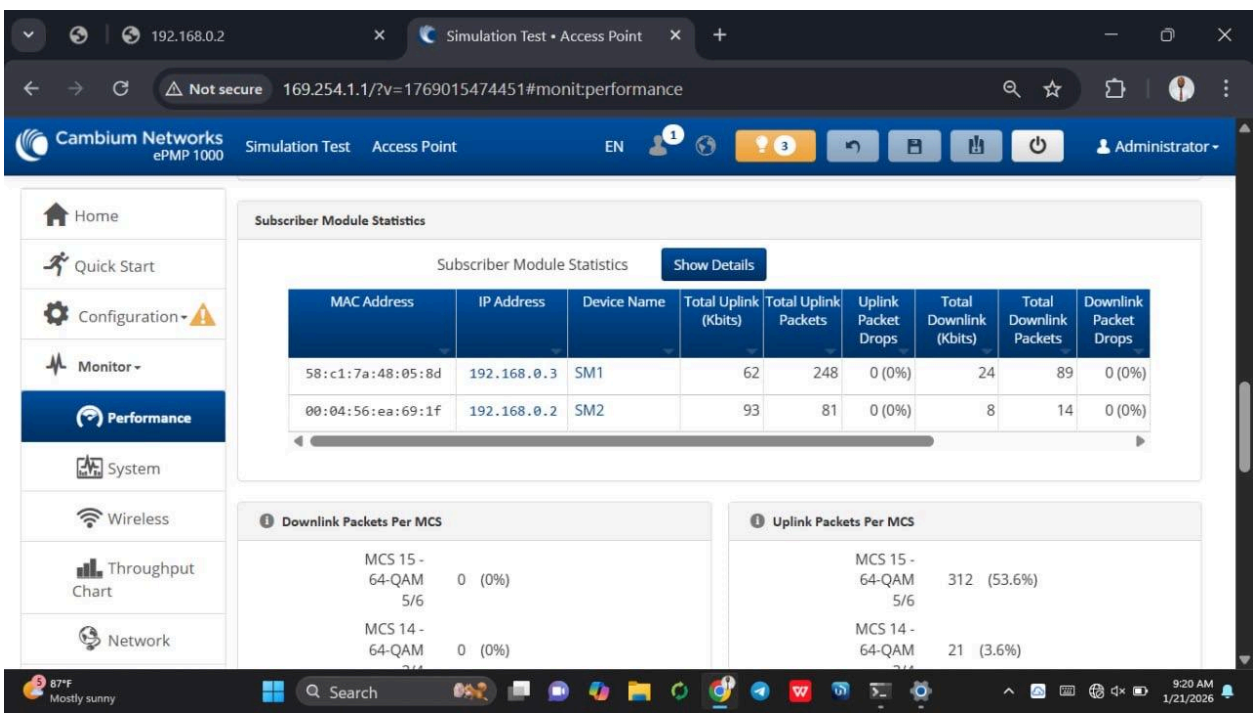

Fig 10: Evidence showing the performance information on the SM Radio Interface

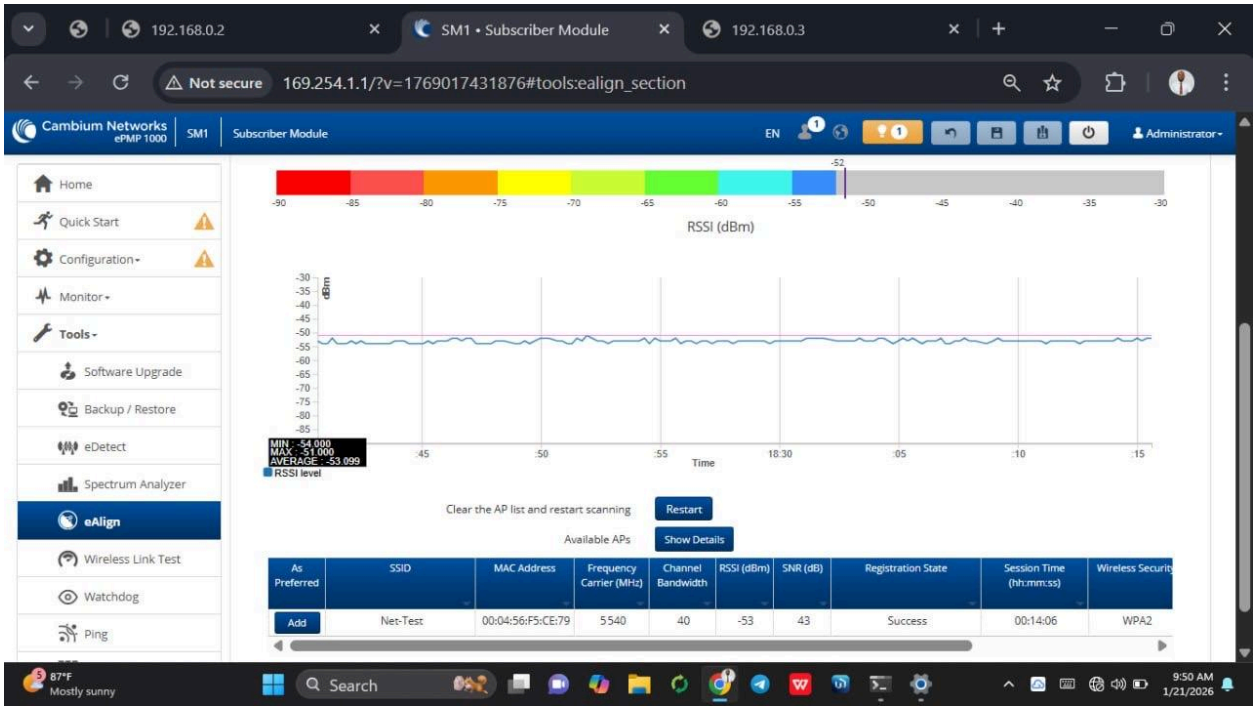

Fig 11: Evidence showing a successful lock of an SM to the AP

Both Subscriber Modules were properly registered on the PtMP base station, confirming correct RF parameters, network addressing, and Layer 3 gateway reachability.

6.0 Main Lab Activities Carried Out

During the lab exercise, the MikroTik router was configured to obtain internet access through a wireless WAN (WWAN) uplink connected to a hotspot. NAT masquerading was implemented to enable private network devices to access external networks via the wireless interface.

A DHCP server was configured to distribute IP addresses within the 192.168.10.0/24 subnet to downstream devices. A Point-to-Multi-Point (PtMP) wireless setup was established, where the Access Point successfully associated with multiple Subscriber Modules.

Routing and gateway reachability were verified to ensure proper traffic forwarding. Connectivity and latency tests were conducted to validate end-to-end communication. The impact of a dual wireless hop architecture was also evaluated.

7.0 Analysis and Results

The MikroTik successfully obtained internet connectivity from the hotspot, and NAT functioned correctly, enabling subscriber access to external networks. The Access Point distributed connectivity effectively, and all devices within the 192.168.0.0/24 network were reachable. Internet access from the subscriber side was confirmed.

However, increased latency and reduced throughput were observed due to the dual wireless hop design. Additionally, the use of a dynamic IP from the hotspot introduced occasional instability.

8.0 Technical Discussion and Engineering Observations

The deployment revealed several important technical observations.

8.1 Wireless WAN Limitations:

The upstream mobile hotspot operates behind Carrier-Grade NAT (CGNAT) and assigns a dynamic public IP address. This introduces limitations such as restricted inbound access, higher jitter, and increased latency, which may affect service stability.

8.2 Double Wireless Hop Impact:

The architecture involves two wireless segments, an hotspot from a Phone connected to MikroTik (Wi-Fi uplink) and AP to SM (PtMP radio link). Each additional wireless hop introduces added latency, reduced throughput, and a higher probability of packet retransmissions due to shared medium contention and signal variability.

8.3 NAT over Radio Access Network

NAT was centrally implemented on the MikroTik router, which functioned as the edge gateway. This provided efficient traffic control and simplified network management. Such a design is suitable for small-scale WISP edge deployments where centralized routing and address translation are required.

8.4 Technical Classification of the Deployment

This laboratory implementation can be technically classified as a **Wireless ISP (WISP) architecture** operating on a **Point-to-Multipoint (PtMP) network model**. The design incorporates both **wireless backhaul and last-mile access**, where upstream connectivity is received wirelessly and redistributed to subscriber nodes.

The deployment also reflects an **edge NAT gateway design**, with the MikroTik router functioning as the boundary device between the private subscriber network and the upstream provider. Furthermore, the structure simulates a simplified **ISP core and access layer**, providing a scaled but architecturally accurate representation of real-world service provider topology.

9.0 Scalability Considerations and Recommendations

For real-world deployment, the lab setup would require infrastructure upgrades to ensure stability and scalability. The **mobile hotspot should be replaced with a fiber ONT** to provide reliable, high-bandwidth internet access. A **licensed microwave backhaul link** would improve throughput and reduce interference compared to a consumer wireless uplink.

An **aggregation switch** should be deployed at the tower to manage multiple subscriber connections efficiently. **VLAN segmentation** is recommended to **separate management, subscriber, and backhaul traffic** for better security and organization. Additionally, **bandwidth management policies** should be implemented to ensure fair usage and maintain service quality as the network scales.

These enhancements would transition the lab model into a more robust and production-ready WISP architecture.

9.1 Real-World Equivalent Architecture

Internet Core (Fiber / Microwave)
↓
Edge Router (MikroTik / Cisco / Juniper)
↓
Tower Aggregation Switch
↓
Cambium Base Station (Sector)
↓ ↓
Customer CPE 1 Customer CPE 2

This lab accurately replicates this model at a smaller scale.

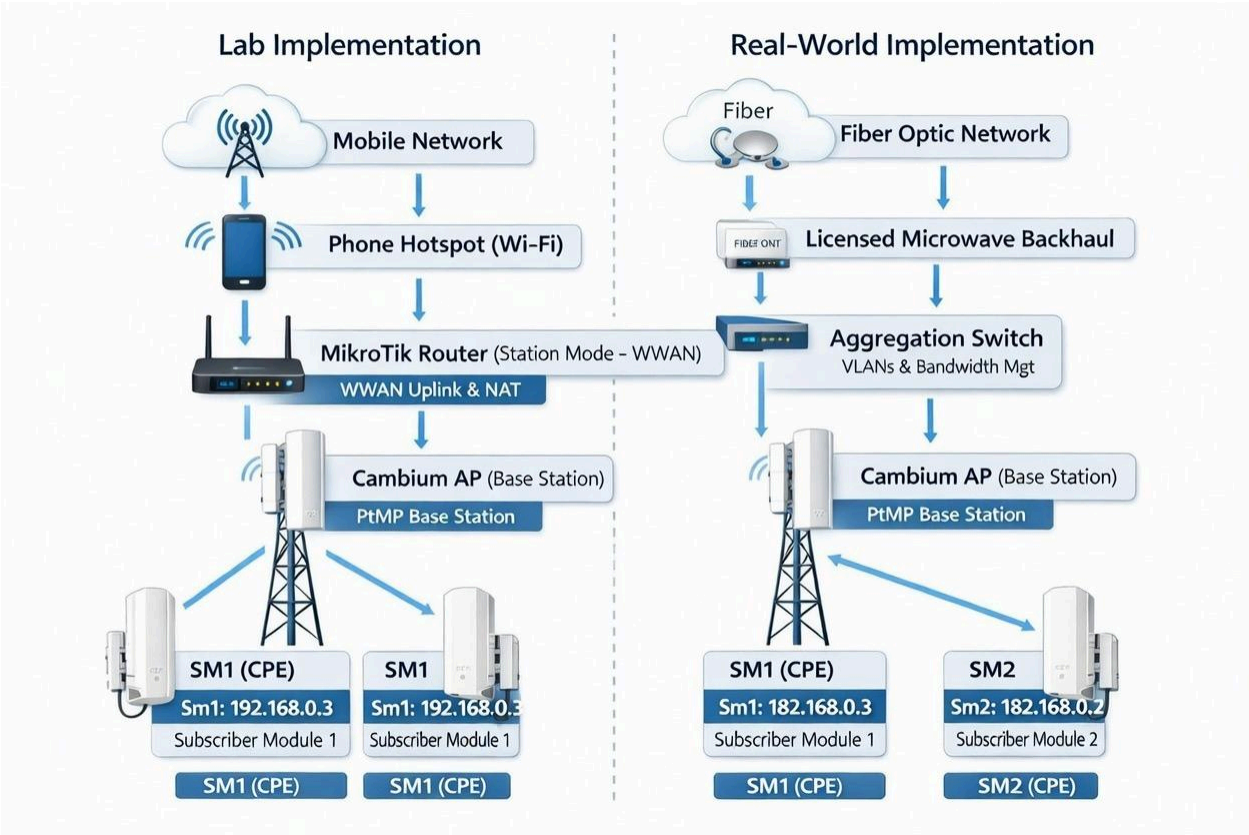

Fig 12: A comparative architectural design of the lab implementation to real-world implementation

10.0 Conclusion

The laboratory exercise successfully demonstrated the deployment of a simulated WISP network using MikroTik and Cambium Point-to-Multi-Point (PtMP) architecture. The implementation achieved successful WWAN configuration, functional NAT and DHCP services, proper AP–SM association, and end-to-end routed internet connectivity.

The exercise also provided a clear understanding of ISP logical layering, particularly in relation to edge routing, address translation, and wireless distribution.

Although the use of a mobile hotspot introduced certain limitations such as CGNAT, increased latency, and occasional instability, the lab effectively modeled a real-world ISP topology. Overall, the implementation represents a scaled-down but architecturally accurate WISP network design and demonstrates practical competency in edge routing and wireless service deployment.